IT4010 Research Project – 2025 July
Topic Assessment Form

Project ID: 25-26J-117

1. Topic (12 words max)

Context-Aware BI Systems: Transition from Static Automation to Intelligent Insight Generation

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Initially, the company managed its client analytics reports through a fully manual workflow. Analysts were required to rebuild the same key performance indicators (KPIs) and graphs from scratch every week, which proved to be inefficient and error-prone. Although core metrics were similar across clients, variations in data structures—such as inconsistent naming conventions or missing fields—meant reports had to be constantly adjusted to ensure accuracy. As the number of clients grew, this manual process became increasingly unsustainable.

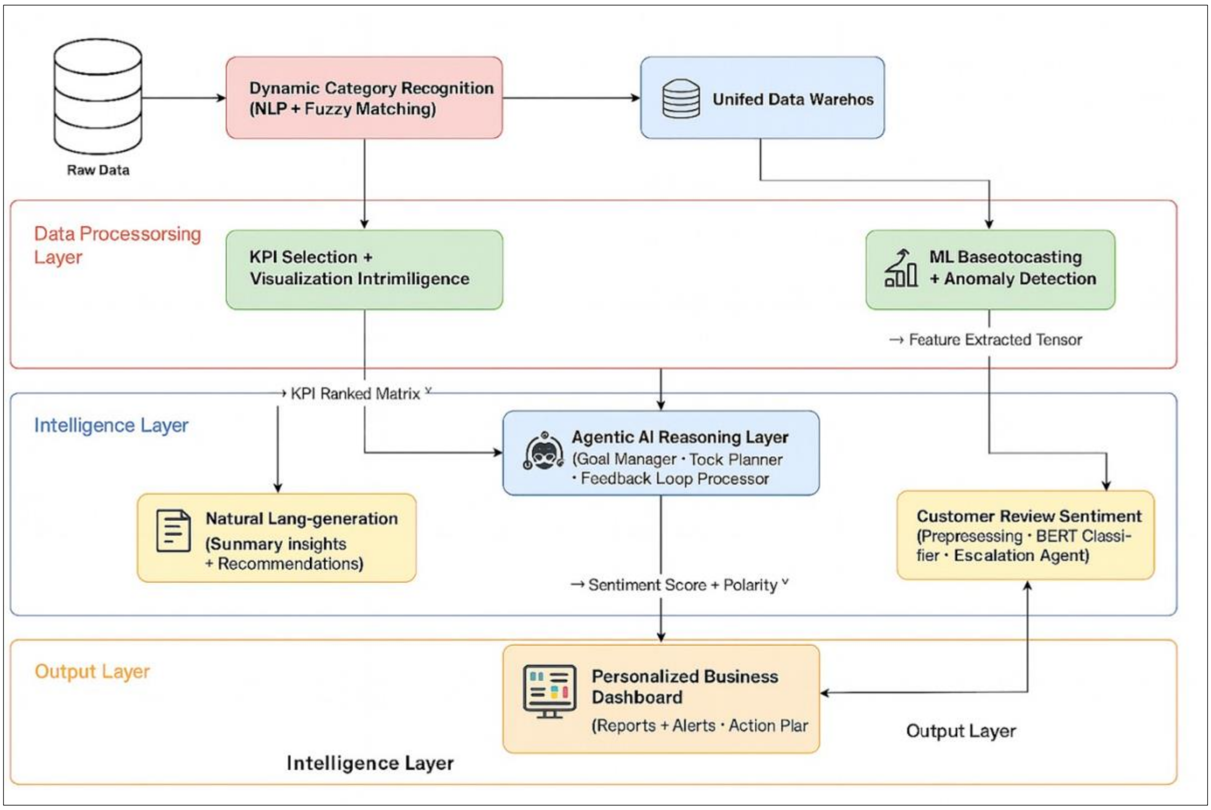
To streamline operations, the company implemented a template-based automation system that generated visualizations and summaries using predefined rules. While this significantly reduced report preparation time, the system lacked flexibility. If the incoming dataset did not exactly match the expected format, the automation either failed or produced incorrect results. Furthermore, the summaries—known as callouts—were static and generic. They failed to capture nuanced trends, unexpected outliers, or client-specific insights because they lacked contextual understanding [1] [2]

Recognizing these limitations, the company began transitioning toward an AI-driven solution capable of contextual reasoning. Researchers have emphasized the growing importance of AI and natural language processing (NLP) in improving business intelligence and decision-making systems [3] , [4]. Rather than relying solely on fixed templates, the new system incorporates adaptive forecasting models, dynamic KPI adjustment, and intelligent insight generation. This evolution aligns with recent studies that show AI's ability to enhance business competitiveness and automate strategic reasoning [5].

Additionally, the integration of NLP-based summary generation allows the system to communicate insights in human-readable form—bridging the gap between data and decision-makers [6]. By learning from feedback and adapting over time, the system transforms traditional reporting into a smart, self-improving business assistant.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The solution is a system that turns the grind of analytics reporting into something quick, smart, and insightful like a trusted assistant who just gets your data. This intelligent solution automates the entire process, making it flexible and valuable for any business. It starts by cleverly sorting your dataset, automatically identifying main and subcategories without manual updates. No more wrestling with inconsistent formats or new data it adapts instantly, saving time and effort. Then, it digs into your data to pinpoint key metrics that matter, spotting patterns to choose the best calculations and visuals. The result is a clear, relevant charts generated in bulk, without the clutter. But it doesn't stop there it explains the graphs like a sharp analyst, using AI to highlight trends, anomalies, and hidden insights. It even predicts future trends by analyzing past sales and factoring in things like holidays. Beyond reports, it uses natural language processing to analyze Google reviews and customer feedback, offering a clear view of sentiment and suggesting targeted marketing ideas. With Agentic AI, it's like having a strategic partner guiding your next move. By blending automation with creativity, this system delivers accurate, actionable insights tailored to any client or dataset, all while keeping things simple and human.


Raw Data
Dynamic Category Recognition (NLP + Fuzzy Matching)
Unifed Data Warehos
Data Processorsing Layer
KPI Selection + Visualization Intrimligence
ML Baseotocasting + Anomaly Detection
→ Feature Extracted Tensor
→ KPI Ranked Matrix
Intelligence Layer
Agentic AI Reasoning Layer (Goal Manager · Tock Planner · Feedback Loop Processor
Natural Lang-generation (Sunmary insights + Recommendations)
Customer Review Sentiment (Prepressessing · BERT Classi-fier · Escalation Agent)
→ Sentiment Score + Polarity
Output Layer
Personalized Business Dashboard (Reports + Alerts · Action Plar
Output Layer
Intelligence Layer

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Making this solution work takes more than just some smart AI system - it requires multiple industry experts that want to turn messy data into clear business insights.

First, you need people who really understand both numbers and business. In industries like retail or food service, this means knowing which metrics matter not tracking everything but also identifying the handful of KPIs that truly show how the business is performing. These analysts bridge the gap between raw data and real-world decisions, understanding how a sudden spike in Tuesday afternoon sales might relate to local events or seasonal trends.

Then come the data engineers who build the plumbing of our system. They create a flexible framework that can take data from different clients and make it all the same. Such as maybe telling a system to recognize that "Beverages", "Drinks", and "Refreshments" are the same thing on different clients' systems.

Additionally, we need machine learning experts. They tell our system to go further than just numbers, to spot patterns a human might miss, predict future trends, and even write natural-sounding insights about what's happening in the data. The system learns to notice things like "Valentine's Day always boosts chocolate sales" or "this product's reviews are getting more negative recently."

The most advanced part of our system can suggest smart next steps - like recommending which customers might respond best to a promotion or flag when it's time to reorder popular items. It’s not just a bot, it’s also a smart business-oriented decision maker.

All of this is built using real client data to really grasps how chaotic client data can be, but system must be aware about the privacy policies that come with dealing with such sensitive data. Every piece of data is anonymized and protected - we're extracting insights, not identities. This real-world testing ground is what lets us build something that works when it meets the complexities of actual businesse

8. Objectives and Novelty

Main Objective

To design a sophisticated, intelligent AI driven reporting system that does more than basic automation, having the potential to understand client data, adapt to different business contexts and finally deliver a full system that aims to provide an organization with personalized, actionable insights by combining AI with human reasoning.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22349910	Categorizing the products (eg: Main/Sub/Coffee or Not Coffee) dynamically using Machine Learning and Agentic AI	**1. Problem Identification and Context** • Identify limitations in the current manual categorization process, which relies on static Google Sheets. • Highlight the scalability issue when managing multiple clients, branches, and evolving menus. • Define the objective: to automate and generalize menu item categorization using AI	**Agentic Categorization Feedback Loop** Uses Agentic AI to initiate human-in-the-loop conversations with branch managers, enabling structured, real-time feedback on ambiguous items and improving classification accuracy. **Context-Aware Categorization Model** Combines historical classifications, vendor-specific menu structures, and known food taxonomies to make intelligent categorization decisions even in the absence of explicit labels.

		and contextual knowledge. **2. Data Collection and Understanding** • Capture client-specific information, including the client name and branch (if applicable). • Collect the full dataset for each client, focusing on unique menu item names. • Identify public sources (e.g., vendor websites) where categorized menu data can be extracted. **3. Automated Data Extraction** • Develop a system that crawls the vendor's website to extract structured menu data. • Match each unique item from the dataset against the extracted online menu. • Use NLP techniques to infer	**Zero-Shot and Few-Shot Learning for New Items** Utilizes advanced NLP models (e.g., embeddings, similarity scoring) to accurately classify unseen or rare menu items without needing prior labeled data. **Human Feedback-Driven Model Refinement** Incorporates manual overrides and human feedback directly into the training pipeline, allowing the system to continuously adapt to client-specific naming patterns and terminology. **Multi-Source Matching and Enrichment** Automatically extracts structured menu data from public vendor websites and aligns it with internal datasets, enriching item context and improving classification precision. **Confidence-Based Categorization with Flagging** Items with low-confidence classification scores are flagged for review instead of being forced into potentially incorrect categories, ensuring data integrity and trust.

		likely categories based on item descriptions and menu sections. **4. Contextual Categorization Logic** • Apply a categorization model that leverages historical categorizations, vendor menu structure, and known food taxonomy. • Automatically assign categories to items using this model. • Flag items with insufficient information or low-confidence predictions. **5. Agentic AI for Human-in-the-Loop Feedback** • Generate a list of uncategorized or ambiguous items for further classification. • Deploy an agentic AI assistant	**Dynamic Knowledge Base Evolution** • Maintains a growing, evolving store of categorization rules — both global (e.g., "Latte → Coffee") and client-specific (e.g., "Big Breakfast → Main") — to improve future performance. **Modular, Scalable Categorization Infrastructure** • Built with a backend architecture that handles multiple client datasets, pipelines, and APIs in a secure and scalable manner, ensuring adaptability to growing data volume and complexity.

		to initiate a structured conversation with the branch manager or business owner. • Collect contextual feedback or direct input on how items should be categorized. **6. Feedback Incorporation and Learning** • Integrate received feedback into the categorization pipeline. • Update the underlying model with new data to improve future performance. • Maintain a dynamic knowledge base of client-specific and general categorization rules. **7. Model Development and Optimization** • Design the classification logic using NLP and ML (e.g.,	

		keyword similarity, BERT embeddings). • Train models to recognize food item patterns across vendors and regions. • Optimize the model using real-world data and continuous feedback loops. **8. System Design and Integration** • Build a modular backend capable of handling multiple client datasets, classification pipelines, and API requests. • Ensure that the architecture supports scalability, automation, and security. • Implement caching and logging to improve response time and traceability. **9. Frontend Interface for Review**	

		- Develop a lightweight interface for internal users or clients to review auto-categorized items. - Allow manual overrides where needed and track changes for model retraining. **10. Testing and Evaluation** - Conduct end-to-end testing across clients to evaluate system performance. - Measure categorization accuracy, fallback rate (items needing human input), and response time. - Refine based on client feedback and operational testing. **11. Deployment and Maintenance** - Deploy the complete system to	

		a secure cloud environment. • Schedule automated data ingestion, categorization runs, and feedback cycles. • Establish monitoring and alerting systems for performance and failures. **12. Documentation and Reporting** • Document the full automation pipeline, models used, and system architecture. • Prepare user documentation for clients and internal teams. • Provide periodic reports detailing improvements in automation rates and feedback incorporation.	

IT22000194	Identifying Optimal KPIs, Aggregations, and Visualizations Dynamically Using Machine Learning	**1. Problem Identification and Context** • Identify inefficiencies in the current manual reporting system, where analysts manually scan and interpret KPIs from spreadsheets. • Highlight scalability challenges when handling multiple clients, large datasets (200K+ rows), and varied reporting needs. • Define the objective: to automate KPI detection, optimal aggregation logic, and visual recommendation using AI. **2. Data Collection and Understanding** • Accept raw data uploads (CSV/Excel) with up to 200k+ rows and 20+ columns from clients. • Perform initial profiling to understand data types (categorical, numerical, datetime) and detect nulls, outliers, and column distributions.	**Agentic KPI Personalization** Engages users in context-aware micro-conversations to refine KPI selection based on business priorities, making the system adaptive instead of rule-based. **Explainable Insight Generation** Each KPI or chart is accompanied by short, human-readable explanations (e.g., "Sales spiked during holiday week"), enhancing transparency and user trust. **Feedback-Driven Learning Loop** Learns from user behavior (accepted, edited, or ignored suggestions) to improve future metric and visualization recommendations — reducing reliance on static rules. **Zero-Configuration Metric Discovery** Automatically identifies key metrics and dimensions from raw, schema-less datasets using semantic inference — no manual tagging or setup required.

		• Identify business-relevant columns (e.g., sales amount, dates, categories, regions) through metadata analysis or optional client input. **3. Automated KPI Extraction** • Use statistical and contextual analysis to detect potential KPIs (e.g., Revenue, AOV, Conversion Rate). • Apply heuristics and templates for industry-specific KPI definitions (e.g., hospitality, e-commerce, healthcare). • Support both absolute KPIs (totals) and derived metrics (percentages, growth, averages). **4. Intelligent Aggregation Engine** • Suggest optimal aggregation strategies for numeric and categorical combinations (e.g., sum revenue by month, count orders per branch). • Use entropy, variance, and other statistical techniques to	

		determine aggregation usefulness. • Handle temporal aggregations (daily, weekly, monthly rollups) with flexibility for time-based comparisons. **5. Chart Type Recommendation System** • Match data patterns and structures with the most appropriate chart types (e.g., pie chart for part-to-whole, line chart for trends). • Incorporate best-practice rules for visual clarity, avoiding common chart misuse. • Rank and recommend top 3 most insightful charts for any metric or table. **6. Human-in-the-Loop (Optional Manual Review)** • Flag low-confidence KPI suggestions or unusual aggregations for human review. • Provide an interface for users to accept, reject, or modify	

		suggested KPIs and charts. • Capture human corrections to improve the system's future suggestions via feedback learning. **7. Learning and Optimization Loop** • Continuously refine recommendations using supervised learning from previous report corrections and client feedback. • Leverage embeddings and semantic similarity to understand column context beyond names (e.g., treating "sales" and "revenue" similarly). • Implement reinforcement loops to update visual and metric selection algorithms. **8. Scalable System Architecture** • Design a backend capable of handling large datasets with streaming or chunked data reading. • Use modular architecture to	

		separate data ingestion, KPI extraction, aggregation logic, and visualization logic. • Ensure performance, fault tolerance, and autoscaling for growing client base. **9. Frontend Interface for Exploration** • Provide an intuitive web-based UI where users can upload data, view auto-generated KPIs, tables, and charts. • Allow editing or reordering of suggested insights before export or download. • Support exporting into PowerPoint, PDF, or embedding into dashboards. **10. Testing and Evaluation** • Perform testing across multiple domains (e.g., restaurant, retail, services) to validate generalization. • Evaluate on metrics like KPI relevance, chart clarity, processing time, and reduction	

		in manual effort. • Conduct user acceptance testing (UAT) with internal teams and early clients. **11. Deployment and Maintenance** • Deploy system on a secure, cloud-native platform with autoscaling and encryption. • Schedule regular model retraining, performance monitoring, and metadata cleanups. • Enable versioning of rules, KPIs, and visual configs for rollback and experimentation. **12. Documentation and Reporting** • Provide clear documentation of automation logic, AI models used, and visual rules applied. • Offer onboarding guides and helpdesk support for internal teams and external clients. • Generate periodic performance reports showing automation % increase, time saved, and accuracy metrics.	

IT22319760	Context-Aware Sales Forecasting and Strategic Planning Agent	**1. Strategic Forecasting Objective Definition** • Identify key business use cases where time-series forecasting adds strategic value (e.g., stock planning, promotional timing). • Frame the objective of moving beyond static dashboards into predictive, planning-oriented intelligence. • Define the KPIs to be forecasted per client: item-level sales, branch-wise performance, etc. **2. Historical Data Analysis and Preparation** • Analyze cleaned sales data (provided post-preprocessing) to understand temporal patterns. • Identify relevant variables such as date, item, branch, seasonality, holidays, promotions. • Engineer additional features that may improve forecasting (e.g., days to next public holiday). **3. Forecasting Model Development**	**Feedback-Driven Forecasting Agent** • Your model learns from past user responses (accepted/ignored suggestions) and improves over time, making it adaptive and semi-autonomous. **Context-Aware Planning** • Integrates public holidays, weekends, and event calendars into forecasting logic, creating more realistic and actionable predictions. **Natural Language-Based Strategy Generation** • Converts numeric predictions

		• Design time-series forecasting models using Prophet, XGBoost, or LSTM. • Train models per item and branch to ensure localized accuracy. • Integrate external event calendars (Poya days, school holidays) into model inputs. • Compare performance of multiple models using RMSE/MAE and choose the best-fit per scenario. **4. Anomaly Detection and Alerting** • Develop anomaly detection pipeline using moving average, Isolation Forest, or Z-score techniques. • Detect drops, spikes, or unexpected flatlines in item/branch sales. • Assign severity scores and confidence levels to flagged anomalies. **5. Agentic AI Integration for Strategic Planning** • Create a logic layer where the forecasting agent makes **decisions** based on predictions. • Example: "If predicted spike > 20% and holiday = True → suggest restock and promo." • Build a small library of goal-driven templates or use GPT-like LLMs for action phrasing.	into human-readable advice using templates or AI, helping business users understand and act quickly. **Explainable Forecasting with Reasoning** • Every prediction includes short explanations (e.g., "due to upcoming holiday trend") and confidence levels to increase trust and clarity.

		6. Natural Language Recommendation Generation • Translate raw forecast values into **text-based suggestions** for business users. • Use NLG techniques to output readable strategies: • "Chocolate cake sales expected to rise 30% next week. Consider increasing supply." • Support multi-lingual generation for Sinhala/Tamil-English hybrid businesses. **7. Contextual Planning with Calendar Intelligence** • Sync forecasting results with real-world dates (public holidays, weekends, religious events). • Refine strategic suggestions based on behavioral patterns (e.g., low traffic on Mondays). • Include upcoming events in planning: • "Plan a discount 3 days before Sinhala New Year." **8. Feedback-Driven Adaptation and Learning** • Log when the business manager accepts/ignores agent advice.	

		• Feed these outcomes back into future prediction models or planning logic. • Enable self-improvement: system prioritizes suggestions that were accepted before. **9. Explainability and Confidence Reporting** • Generate short explanations for why the model made a prediction (XAI). • "Sales prediction influenced by previous Poya spike + Saturday trend." • Include confidence intervals or probability of prediction failure in the output. **10. Dashboard Integration and Visualization** • Forward prediction values, alerts, and recommendations to the shared dashboard pipeline. • Highlight forecasted metrics, anomaly alerts, and agent-suggested actions. • Work with frontend team to visualize forecast confidence and event overlays. **11. Testing and Evaluation** • Conduct testing using historical data and simulate future weeks.	

		• Measure accuracy, anomaly detection recall, and recommendation usefulness. • Collect qualitative feedback from beta testers or mock business users.	
IT22323552	Enhance customer review analysis using AI to improve sentiment detection, quantify emotional intensity, and identify contextual triggers, while generating personalized response templates and prioritizing critical issues for faster business action.	1. **Literature Review and Problem Definition** • Review existing sentiment analysis research, focusing on BERT and emotion detection in customer reviews. • Identify challenges: limited emotional granularity, lack of contextual triggers, and generic response systems. • Highlight gaps where contextual emotion intensity analysis can enhance customer engagement. 2. **Selection of AI Models** • Evaluate NLP models (e.g., BERT, RoBERTa) and topic modeling (e.g., LDA) for	**Limited Emotional Granularity:** • Current sentiment analysis often sticks to basic positive, negative, or neutral classifications. Fine-grained analysis, such as detecting emotion intensity (e.g., mild vs. extreme anger), is underexplored. **Sarcasm and Irony Detection**: • Challenge of detecting sarcasm and irony, which often misalign with literal sentiment. Therefore further integrating sarcasm detection could enhance accuracy.

		sentiment and intensity analysis. • Choose the best model based on evaluation for sentiment and intensity classification, paired with LDA for contextual trigger extraction. • Consult superiors to validate model choices for real-time review processing. 3. **Data Collection and Preprocessing** • Gather review datasets (e.g., google) with labeled sentiments and custom emotion-intensity annotations. • Clean data to remove noise (e.g., irrelevant text, duplicates) and standardize formats. • Prepare datasets for training BERT and LDA models, ensuring compatibility with intensity scoring.	**Data Quality and Domain Shift:** • Imbalanced datasets and domain specific performance limits the models generalizability, therefore ability to generalize it will outsmart the already existing ones.

		4. **Implementation of AI Models** • Develop BERT prototype for sentiment and emotion intensity classification, plus LDA for theme detection and Agentic AI to respond or escalate based on analysis. • Train models on prepared datasets, optimizing hyperparameters for accuracy. • Validate using F1-score for sentiment and mean squared error for intensity prediction. 5. **Integration into Web Application** • Design architecture to integrate BERT and LDA models into a web-based system. • Build APIs for real-time sentiment analysis, intensity scoring, and response generation.	

		• Ensure scalability for high-volume review processing. 6. **Frontend Development** • Create an intuitive dashboard to display sentiment, emotion intensity, and review themes. • Add features for visualizing priority scores and response suggestions. • Conduct usability testing with business users to refine the interface. 7. **Backend Development** • Build infrastructure for data storage, model inference, and user authentication. • Implement secure APIs to connect frontend with model outputs.	

		- Ensure system reliability for continuous review processing. 8. **Testing and Evaluation** - Perform unit, integration, and performance testing of the system. - Evaluate model accuracy against benchmarks and real-world review data. - Refine models and UI based on feedback and metrics like response relevance. 9. **Deployment and User Training** - Set up cloud-based deployment for the web application. - Train business users on interpreting dashboards and acting on insights.	

		• Monitor system performance post-deployment. 10.**Documentation and Final Report** • Document methodology, model development, and system integration. • Summarize findings, including how emotion intensity improves outcomes. • Present results to stakeholders, highlighting data science innovations.	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Wagoda Pathirage V K **IT22349910**	Dynamic Category Recognition leverages NLP to automatically identify and extract menu categories, then applies fuzzy matching to align them with industry-standard taxonomies. For edge cases, an agentic AI system steps in to resolve discrepancies. This aligns with our degree by exploring Data Engineering areas where we build and scale data pipelines, while the Machine Learning model trains and optimizes the classification
ADIKARI A I **IT22000194**	KPI & Visualization Intelligence uses statistical analysis to determine the most relevant KPIs, then recommends the best graph types to visualize the data, analyzing which metrics matter most, suggesting the best charts to visualize them, and summarizing data effectively. It also employs adaptive aggregation logic to dynamically adjust how information is grouped and displayed. This component aligns with our degree by focusing. The strong focuses here are on students' ability to analyze data exploring the importance of data scientist skills of a student.
Vinushan V **IT22319760**	The component is designed to analyze historical business data and produce meaningful, forward-looking insights that support strategic decision-making. By applying models such as Prophet and XGBoost, the system predicts future sales trends across different branches and products. To ensure these predictions are realistic and actionable, the system integrates contextual data — such as holidays, weekends, and local events — into its forecasting pipeline. In addition, this component translates numeric predictions into natural-language strategic recommendations using rule-based logic or generative AI techniques. It also includes a feedback loop that allows the system to learn from user responses, enabling it to refine future suggestions over time. This approach reflects the core values of applied data science: turning complex

	data into interpretable, real-world insights that drive intelligent business actions.
Anupama S K D N **IT22323552**	Review & Sentiment Management uses advanced NLP techniques, like BERT-based sentiment analysis, to classify customer reviews and feedback

	as positive, negative, or neutral, uncovering actionable insights. It automatically generates tailored response templates to streamline customer engagement and assigns priority scores to flag urgent issues. This connects to data science through the NLP Specialist's work in fine-tuning language models for accurate sentiment detection, a core machine learning task. This process ties directly to data science by combining machine learning, data visualization, and business intelligence to transform unstructured review data into strategic outcomes.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Jenny	Krishara	Jenny 2025/06/26
Co-Supervisor	Dr	Dinuka	Wijendra	26/06/2025
External Supervisor				

Summary of external supervisor's (if any) experience and expertise

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	**Signature**

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

References

[1] F. J. M. a. M. Oliveira, "Natural language processing in business intelligence: A review," in *Proceedings of the International Conference on Artificial Intelligence and Advanced Manufacturing*, 2023.

[2] A. R. a. M. Sharma, "Artificial intelligence in strategic business decisions: Enhancing market competitiveness," *International Journal of Business Research,* vol. 14, no. 2, pp. 55-68, 2024.

[3] L. C. a. D. Zhang, "AI in market research: Transformative customer insights – A systematic review," *International Journal of Marketing and Research Excellence,* vol. 11, no. 1, pp. 24-39, 2024.

[4] B. C. Studies, " What is AI for monitoring business KPIs?," 2024. [Online]. Available: https://businesscasestudies.co.uk/what-is-ai-for-monitoring-business-kpis/.

[5] A. S. a. R. Kumar, "Intelligent automation of graphical reporting systems," *IEEE Access,* vol. 12, p. 11034–11045, 2024.

[6] M. Patel, "What AI means for business reporting," *International Journal of Management Research and Engineering (IJMRE),* vol. 16, no. 1, pp. 1-17, 2024.